PAULINA CZARNOTA

Final-Year Computer Science Student | Software Development | Testing | Cybersecurity
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PROFESSIONAL PROFILE

Final-year Computer Science student at TU Dublin (graduating October 2027) with strong Java and object-oriented programming skills, built through Algorithms and Data Structures, Software Engineering and a Java EE web application using Servlets, JSP, JDBC, MySQL and the MVC pattern. Solid grounding in SQL and databases (PostgreSQL, MySQL, Oracle), UML and design patterns, unit testing, Git and CI/CD pipelines with GitHub Actions and Jenkins. Completed a Mobile Network Testing internship at 1GLOBAL, testing APIs, tracing defects to root cause and working in an Agile team. Led a student team on a real-client project, from stakeholder requirements to documentation, presentations, React implementation and end user testing.

PROFESSIONAL EXPERIENCE

Mobile Network Testing Intern | 1GLOBAL
May 2026 to Nov 2026 | Pembroke House, Dublin 2 | MVNO/MVNE global telecommunications platform

- Contributed to the pre-launch validation of Lidl Connect Ireland (October 2026), running functional, regression, exploratory, interoperability, B2B and UX tests on physical iOS and Android devices and the Lidl staging app.
- Tested SIM and eSIM services (provisioning, QR activation, eSIM swaps, APN, roaming, port-in/port-out) and validated VoLTE, VoWiFi, SMS, data, voicemail and emergency services across Lidl Connect, Monzo Mobile, Revolut Mobile, Tesco Mobile, Eir, Vodafone and Three.
- Tested provisioning APIs in Postman (webhooks, CDRs, data draining, Magnolia flows), validating JSON responses, and used Grafana to monitor network and provisioning activity and trace faults to root cause.
- Logged and retested defects in Jira, documented results in Confluence and Excel, and worked with QA, software and network engineers in an Agile team.

EDUCATION

BSc (Hons) Computer Science (TU856) | Technological University Dublin
2021 to 2027 (Expected) | Grangegorman, Dublin 7

- Final-Year Modules: Advanced Databases; Forensics; Advanced Security; Machine Learning for Data Analytics.
- Previous Coursework: Algorithms and Data Structures (Java); Object-Oriented Programming (Java and Python); Software Engineering; Software Testing; Databases; Web and Client-Server Programming; Mobile Application Development; Cloud Computing and DevOps; Operating Systems; C Programming; Data Communications; Cryptography and Cyber Security; Artificial Intelligence; Universal Design and Assistive ICT.

Leaving Certificate | Mount Carmel Secondary School, Dublin
2015 to 2021

- 480 CAO points. Higher Level: History (H3), Business (H3), Home Economics (H3), Polish (H3), French (H4). Ordinary Level: English (O1), Maths (O1).

KEY PROJECTS

Task Management Web Application | Java EE, Servlets, JSP, JDBC, MySQL, MVC
GitHub

- Designed and built a web application with full CRUD for tasks, using a Task model, a controller servlet, a DAO layer and a JDBC utility class, with a class diagram, four sequence diagrams and a presentation.

Graph Algorithms and Data Structures | Java
GitHub

- Implemented Prim's, Kruskal's and Dijkstra's algorithms and graph traversals from scratch, with a report on design and complexity.

iPlanit Accessible Action Plan Redesign (Team Lead) | Co-Design, React, WCAG 2.1
GitHub

- Led a student team with St. John of God Community Services: ran stakeholder meetings and co-design workshops, gathered requirements, and wrote the project charter, risk plan, design document, test plan and user manual.
- Contributed to the React implementation, supported teammates with coding issues, ran end user testing and presented at all four milestones.

Software Testing and Usability Testing | Python, pytest, unittest, coverage, pylint
GitHub | GitHub (Labs)

- Built a unit test suite for a music-library API (search, filtering, invalid input, 200ms performance target) with coverage and pylint reports; practised TDD, black box and white box testing, debugging and a formal usability study.

Databases and SQL | PostgreSQL, MySQL, MongoDB, Neo4j, Python
GitHub

- Wrote advanced SQL with window functions, triggers, normalisation to 3NF and B-tree indexes in PostgreSQL, and connected PostgreSQL to Python with psycopg2 and an external REST API.

Web Applications with Database Backends | Node.js, Express, PHP, MySQL
GitHub | *GitHub (Library Site)*

- Built a Node.js and Express search application backed by MySQL, and a PHP and MySQL library reservation site with registration, login, validation, sessions and search.

DevOps and CI/CD Labs | Jenkins, GitHub Actions, Docker, Kubernetes, Vagrant
GitHub (Jenkins) | *GitHub (Actions CI)*

- Built a Jenkins pipeline and a GitHub Actions workflow running unit tests on every push, and deployed a Flask app with Docker and Kubernetes.

Android Fitness Tracker Application | Kotlin, Jetpack Compose, MVVM, Room
GitHub

- Android app using MVVM and Room/SQLite, with biometric login and Android Keystore and PBKDF2 security.

TECHNICAL SKILLS

Programming Java; object-oriented programming; Python; C; Kotlin; JavaScript; SQL; PL/pgSQL; PHP; Bash.

Backend & APIs Java EE (Servlets, JSP, JDBC); MVC and DAO patterns; Node.js and Express; RESTful API testing (Postman, JSON); client-server programming (HTTP, TCP sockets).

Databases PostgreSQL; MySQL; Oracle SQL; MongoDB; Neo4j; query optimisation and indexing; normalisation; triggers; data modelling and ER design.

Testing & Debugging Unit and integration testing (JUnit, unittest, pytest); TDD; black box and white box testing; coverage; debugging; defect reporting (Jira); root-cause analysis; Postman.

DevOps & Tools Git and GitHub; GitHub Actions; Jenkins; Docker; Kubernetes (basics); Vagrant; Prometheus and Grafana; Linux; Eclipse; VS Code; PyCharm; Confluence.

Design & Methods UML (class, sequence, use case); design patterns (Singleton, Composite, Observer, Adapter); layered architecture; Design by Contract; Agile and Scrum; requirements gathering; technical documentation.

Security Secure coding; password hashing; Android Keystore; AES and RSA; vulnerability management.

Professional Problem-solving; troubleshooting; clear communication; teamwork and leadership; ownership of work.

CERTIFICATIONS & LANGUAGES

- Udemy: Cybersecurity Vulnerability Management; Computer/Digital Forensics; The Web Developer Bootcamp 2024; The AI Engineer Course 2025: Complete AI Engineer Bootcamp; PowerShell File and Folder Management.
- TU Dublin: Certificate for Participation in Co-Design Team Projects (2024 to 2025), signed by the Head of School.
- Languages: English (full professional proficiency); Polish (native/bilingual).

ADDITIONAL EXPERIENCE & VOLUNTEERING

STEPS Engineering Your Future Programme | ESB, Dublin 3
Mar 2019

- Placement at ESB headquarters alongside electrical, mechanical, civil, chemical, biomedical and CAD engineers, with lab work, calculations and team problem-solving.

Childcare Worker | Pitter Patter Community Crèche & Pre-School, Dublin 1
Mar 2019

- Supervised children aged 1 to 5, ran daily routines and led educational and play sessions, handling issues calmly in a fast-paced setting.

Retail Sales and Deli Assistant | Eurospar, Dublin 11
Nov 2018

- Customer service and fresh food preparation to HACCP standards, with stock rotation, replenishment and promotional displays.

Step-In Experience Programme | A&L Goodbody LLP, Dublin 1
Oct 2018

- Selected for a competitive programme at a leading Irish law firm, shadowing legal professionals and joining workshops and case studies.

Volunteer | Friends of the Elderly Ireland, Dublin 1
Sep 2018 to Jan 2019

- Organised and led weekly social events and clubs for older adults.

Fundraising Volunteer | Oesophageal Cancer Fund (OCF)
Mar 2019

- Fundraising to raise money and awareness for an Irish health charity.